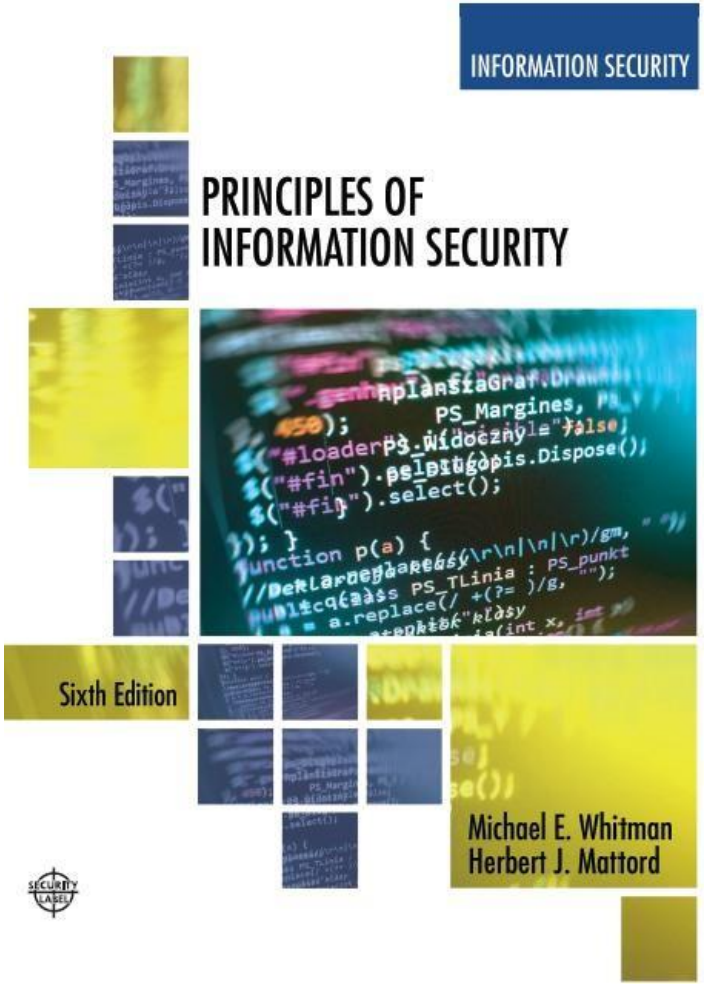
Principles of Information Security

Sixth Edition



Chapter 2

Security Technologies:

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firewall

What is a firewall?

A software and hardware that provides isolation system between an organization's internal network and other networks, it uses two of mechanisms: one mechanism to block traffic, and the other mechanism to permit traffic .

It may be a hardware device (see Figure 1) or a software program (see Figure 2) running on a secure host computer.

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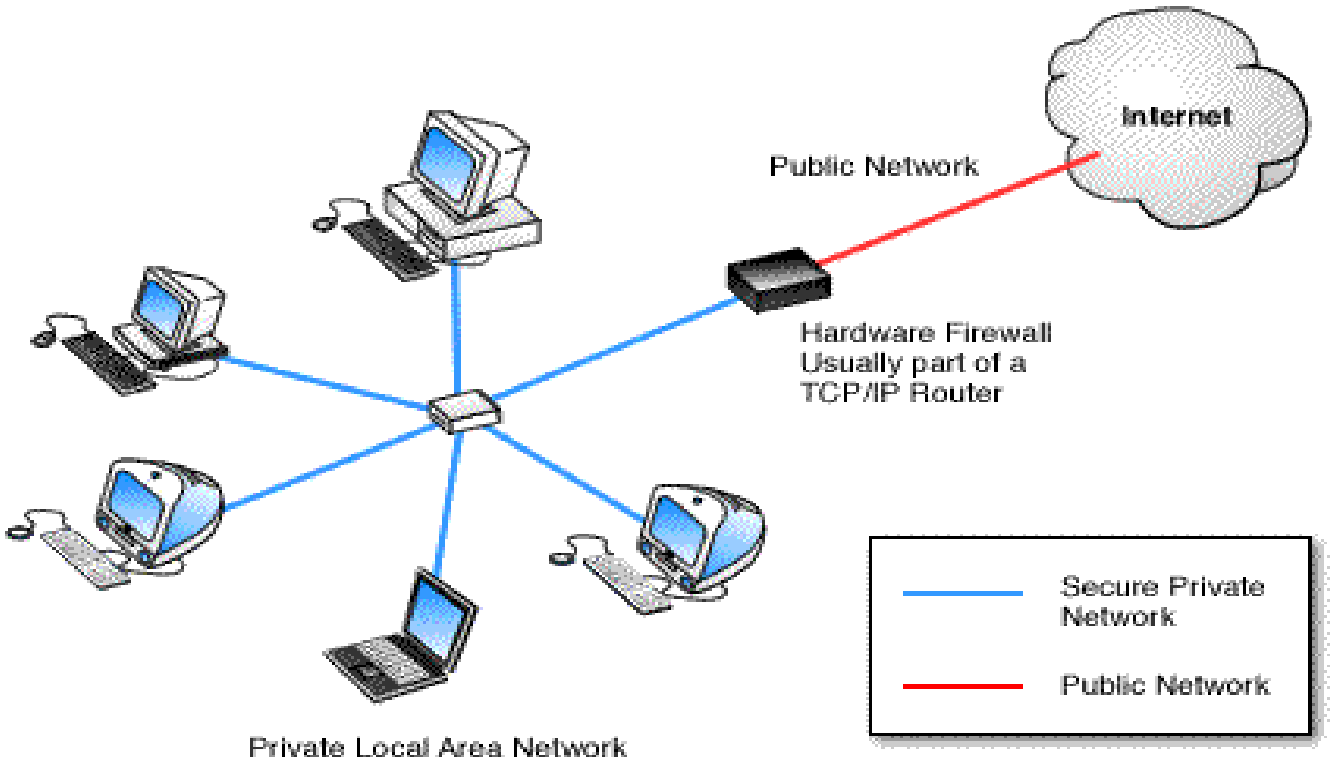
Figure 1: Hardware Firewall

- Hardware firewall providing protection to a Local Network



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Hardware Firewall

- **Hardware:** A hardware firewall can be a router or any other security product that is placed between your PC and the outside world.

- More than one PC can be connected with that security device or router, and each will be protected by the firewall.

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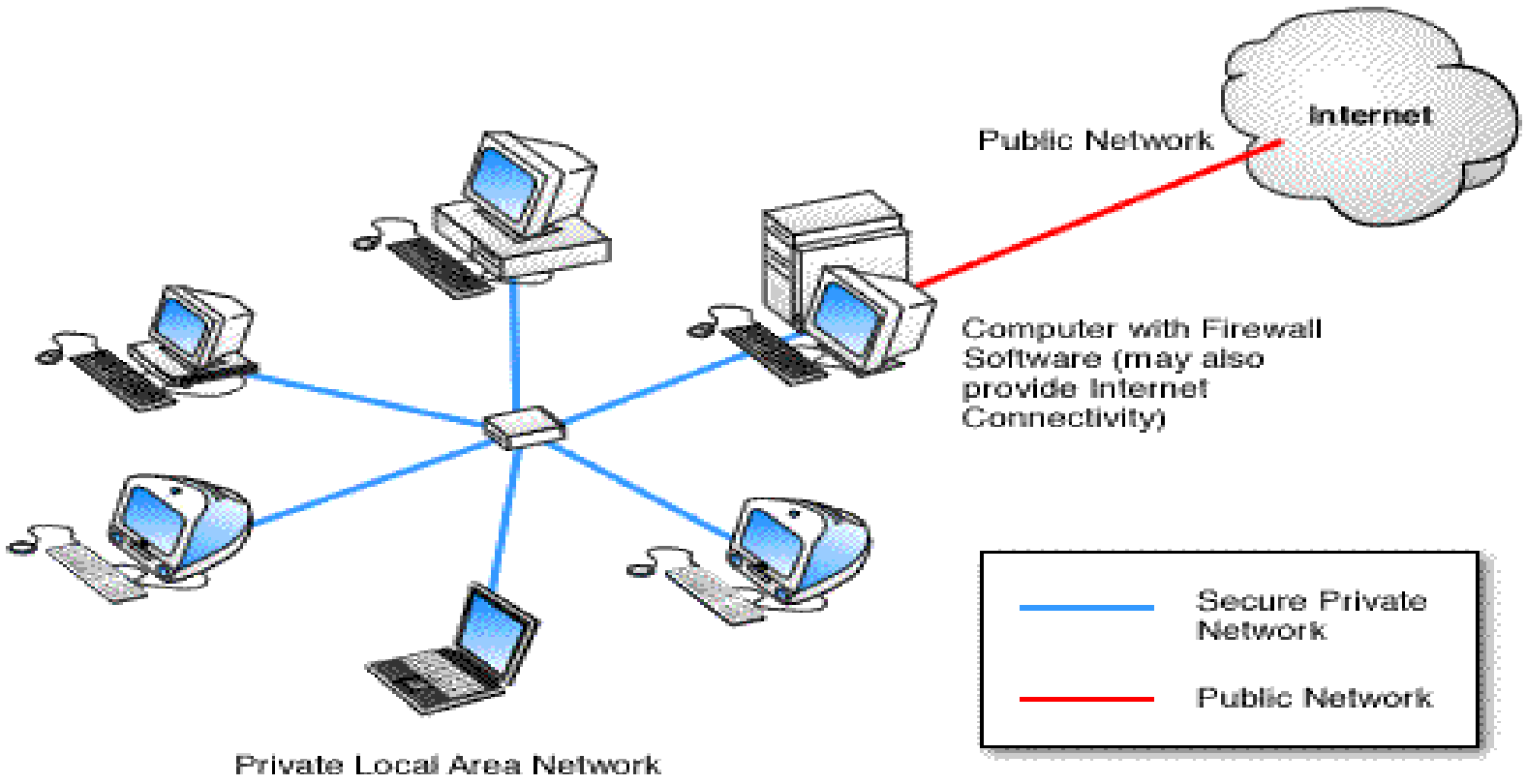
Figure 2: Computer with Firewall Software

- Computer running firewall software to provide protection



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Firewall Software

- **Software:** A software firewall is installed on your PC, it will deflect incoming attacks and hide open ports.
- It will warn you about suspicious outgoing traffic.
- You can have a software firewall on each computer for strong protection.
- By installing software firewall, an attacker will have to cross two level security to access your PC.

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Features and functions of Firewalls

- Enforces the access control policy.
  - Close down a port completely.
  - Open a port to only specific IP addresses.
  - Block (blacklist) specific IP addresses.
  - Open ports only during specific time frames.

- Only transit point between networks. (all traffic flows through the firewall).

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Features and functions of Firewalls

- Many firewalls include VPN services as well, allowing encrypted communication with designated users from outside of the local area network.
- A firewall helps to protect the internal network by hiding names of internal systems from the external hosts.
- It prohibits potentially vulnerable services from entering or leaving the network.
- Audits and alarms can be implemented on the firewall systems.

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Types of Firewalls.

There are different types of firewall techniques that will prevent unauthorized individuals to access your crucial data over the internet:

- **Proxy Server.**

Proxy server checks all messages entering and leaving the network and hides the true network addresses.

- **Circuit-level Gateway.**

Circuit-level gateway is used for TCP or UDP connection. On establishment of secure TCP or UDP connection, data packets can flow between the hosts without further checking.

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Types of Firewalls.

- **Packet Filter.**

Packet filter firewall checks each data packet entering or leaving the network. Packet filtering is fairly effective and transparent to users, but it is difficult to configure.

- **Application Gateway.**

Application Gateway security mechanisms is applied to specific applications, such as Telnet and FTP servers. It secures server effectively but slow down performance.

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Types of Firewalls.

- **Next Generation Firewall (NGFW) الاسم كامل**

NGFW – next generation firewall is the newer class of firewalls that filters internet and network traffic based upon traffic types using specific ports. Next Generation Firewalls (NGFWs) feature the basic functionalities of a standard firewall with quality of service (QoS) in order to provide deeper and smarter inspection.

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Access Control

- **Access control** is the protection of information resources or services from access or use by unauthorized entities (organizations, people, machines, processes).

- **There are two types of access control:**

1. Physical access control.

   limits access to campuses, buildings, rooms and physical IT assets.

1. Logical access control.

   limits connections to computer networks, system files and data.

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Access Controls

- **In general**, all access control approaches rely on the following four mechanisms, which represent the four fundamental functions of access control systems:

  1. **Identification**: I am a user of the system.
  2. **Authentication**: I can prove I'm a user of the system.
  3. **Authorization**: Here's what I can do with the system.
  4. **Accountability**: You can track and monitor my use of the system.

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Access Controls

1. Identification

- **Identification:** The access control mechanism that requires the validation and verification of an unauthenticated entity's purported identity.
  - Identifiers can be composite identifiers, concatenating elements—department codes, random numbers, or special characters—to make them unique.
  - Most organizations use a single piece of unique information, such as a complete name or the user's first initial and surname.

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Access Controls

2. **Authentication**

- **Authentication:** The access control mechanism that requires the validation and verification of an unauthenticated entity's purported identity.
- Authentication factors
  1. Something you know
     - Password: a private word or a combination of characters that only the user should know
     - Passphrase: a series of characters, typically longer than a password, from which a virtual password is derived

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Access Controls

2. Something you have
   - Dumb card: ID or ATM card with magnetic stripe
   - Smart card: contains a computer chip that can verify and validate information
   - Synchronous tokens
   - Asynchronous tokens
3. Something you are
   - Relies upon individual characteristics
   - Strong authentication

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Access Controls

3. **Authorization**

- **Authorization**: The access control mechanism that represents the matching of an authenticated entity to a list of information assets and corresponding access levels.

- Authorization can be handled in one of three ways:
  - Authorization for each authenticated user
  - Authorization for members of a group
  - Authorization across multiple systems

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Access Controls

4. **Accountability** نقط

- **Accountability**: The access control mechanism that ensures all actions on a system—authorized or unauthorized—can be attributed to an authenticated identity. Also known as auditability.
- Accountability is most often accomplished by means of system logs and database journals, and the auditing of these records.
- Systems logs record specific information.
- Logs have many uses.

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Biometrics

- Approach based on the use of measurable human characteristics/traits to authenticate identity.
- Only fingerprints, retina of eye, and iris of eye and DNA are considered truly unique.
- Evaluated on false reject rate, false accept rate, and crossover error rate.
- Highly reliable/effective biometric systems are often considered intrusive by users.

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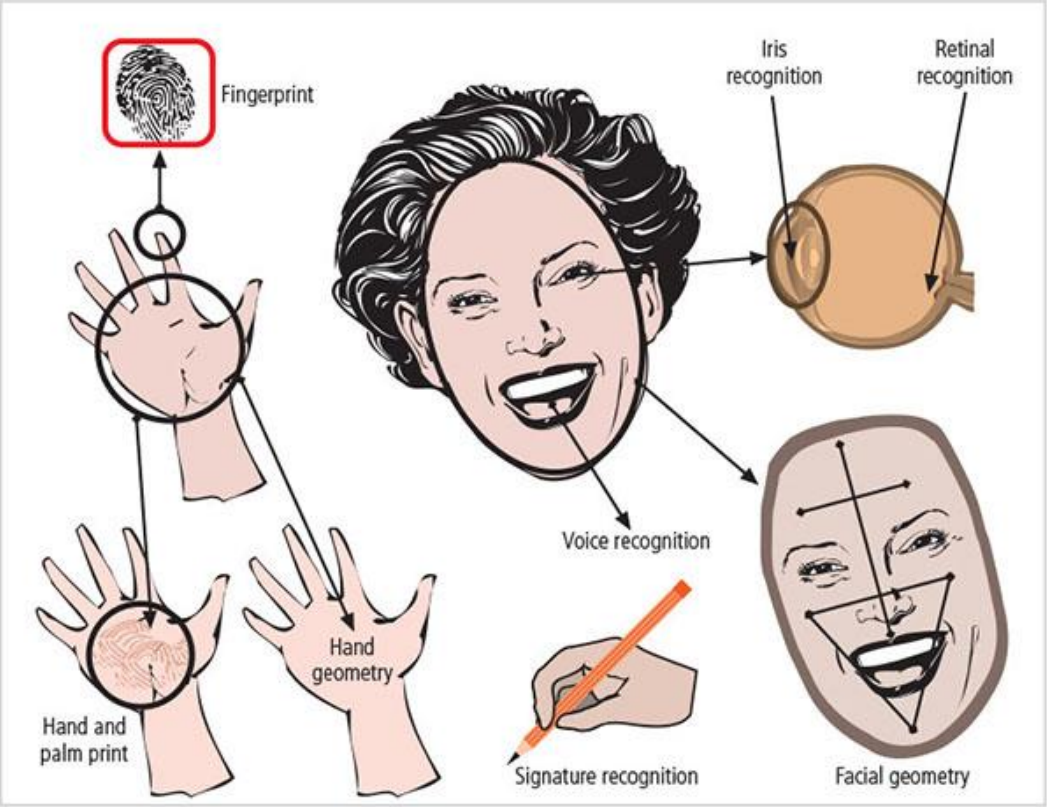
Figure 6-5 Biometric recognition characteristics



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What Is A VPN?

- A VPN (Virtual Private Network) is a private network that protects the online privacy of users who surf the internet. And not only from hackers and other malicious attacks – it even keeps user data safe from governments and their own Internet Service Provider (ISP).

- VPN stands for Virtual Private Network. It encrypts your connection over a public or unsecured network so others can’t see what websites you’re accessing or where you are.

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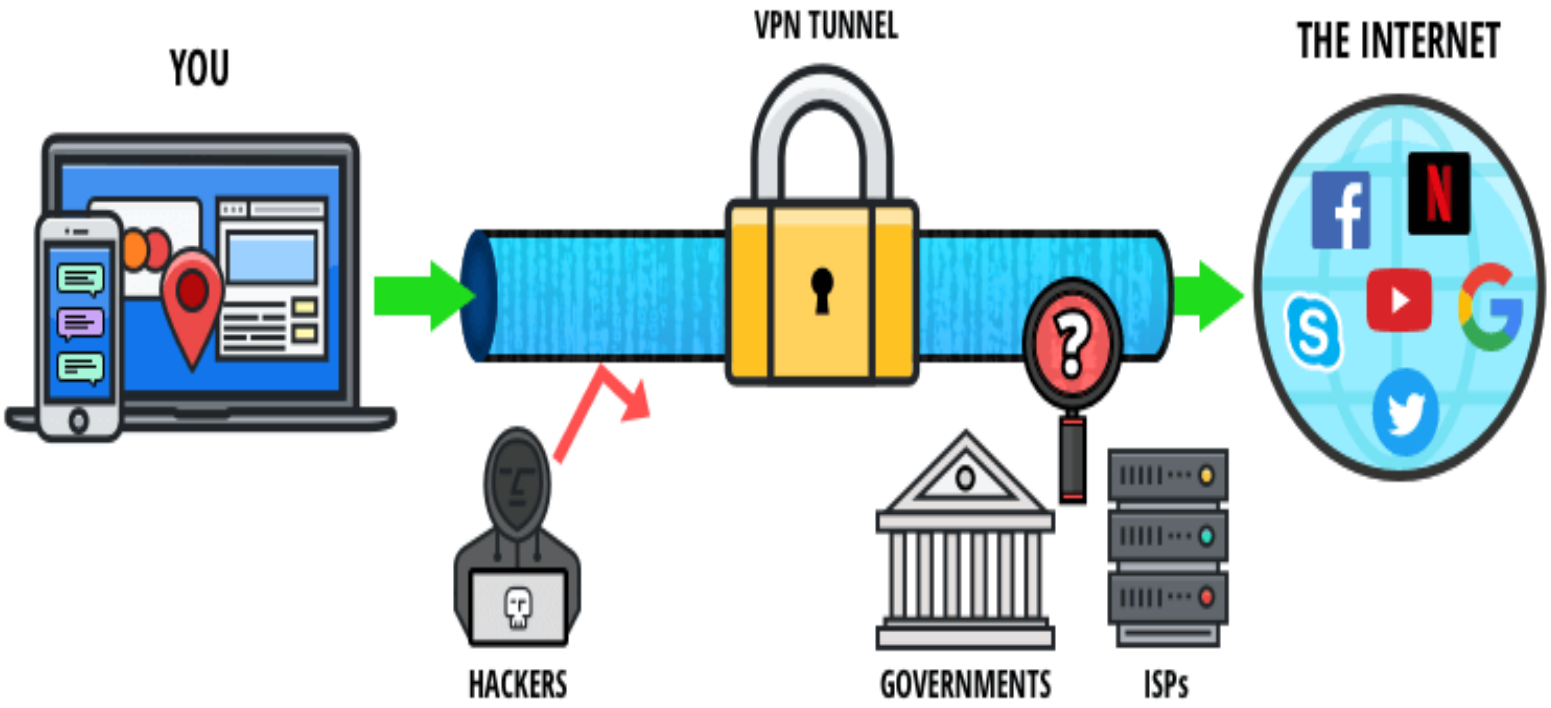
What Is A VPN?



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Is VPN traffic encrypted?

- Yes, traffic on the virtual network is sent securely by establishing an encrypted connection across the Internet known as a tunnel. VPN traffic from a device such as a computer, tablet, or smartphone is encrypted as it travels through this tunnel. Offsite employees can then use the virtual network to access the corporate network.

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Features of the VPN?

- **Hide your IP address**

  Connecting to a Virtual Private Network often conceals your real IP address.

- **Change your IP address**

  Using a VPN will almost certainly result in getting a different IP address.

- **Encrypt data transfers**

  A Virtual Private Network will protect the data you transfer over public WiFi.

- **Mask your location**

  With a Virtual Private Network, users can choose the country of origin for their Internet connection.

- **Access blocked websites**

  Get around website blocked by governments with a VPN.

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Virtual Private Network Security

- Security is the main reason why corporations have used VPNs for years. VPNs use advanced encryption protocols and secure tunneling techniques to encapsulate all online data transfers.

❖ Note that:

- the difference between the technologies firewall and VPN is that a firewall protects your data while on the computer and a VPN protects your data on the web.

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Virtual Private Network Protocols

A virtual private network is created by establishing a virtual tunnel between two endpoints via a virtual tunneling protocol or by data encryption. Some of the most popular VPN protocols include **IPsec, SSL/TLS, PPTP and L2TP**.

- **PPTP - Point-to-Point Tunneling Protocol** is the oldest VPN protocol developed by a consortium found by Microsoft, which is supported by vast majority of operating systems. The encryption based on 128-bit key has been cracked, and it is no longer considered very secure.

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Virtual Private Network Protocols

- **L2TP/IPsec - Layer 2 Tunnel Protocol with IPsec** encryption based VPN provides more secure service with more features than PPTP. L2TP uses UDP port 500, so advanced configuration may be required to open NAT firewall.

- **Open VPN** - is the open source technology developed on Open **SSL/TLS** (Secure Sockets Layer) / (Transport Layer Security), which provides very secure connection and strong encryption between a web server and a browser. It has become the default VPN connection type, and is widely supported by 3rd-party software including iOS and Android.

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